

Determination of critical field and demagnetization factor of type-I superconductor lead

Hrvoje Krizic and Simon Zehnder
ETH Zurich, CH-8093 Zurich, Switzerland
Laboratory for Muon Spin Spectroscopy,
Paul Scherrer Institute,
CH-5232 Villigen PSI, Switzerland
 (Dated: October 3, 2026)

The use of the muon spin rotation/relaxation (μ SR) technique to study type-I superconductivity is discussed. At very low temperatures, a superconductor expels external magnetic fields from its interior, resulting in a perfect diamagnetism (Meissner State). By applying higher external fields, the superconductor eventually transitions to a nonsuperconducting state at a critical field B_c . In addition to both states, a mixed state (intermediate state) occurs between them. The μ SR technique allows one to accurately determine the value of B_c . The technique also allows for the precise measurement of the volume of the sample in both the normal and superconducting states. Measurements are performed on a disk shaped Pb sample. The thermodynamic critical field $B_c = (77.16 \pm 0.06)$ mT was determined, which is in very good agreement with literature. Furthermore, the demagnetization factor $N = (0.71 \pm 0.02)$ of the lead disk was calculated from the boundaries of the intermediate state.

I. INTRODUCTION

Superconductivity describes the phenomenon of zero electrical resistivity of a material. This phenomenon occurs when the material, called the superconductor, is cooled down below a certain critical temperature T_c [1]. At this extreme low temperature, the external magnetic fields are expelled from the material [2]. This condition is called the Meissner state. This state not only breaks down for high temperatures, but also if one increases the external magnetic field. Above a critical magnetic field B_c the superconducting state transitions to a nonsuperconducting state.

At the transition from the superconducting state to the normal state, there can be an abrupt breakdown of the Meissner state which we call type-I superconductors [3]. In type-II superconductors there will be a mixed state between the superconducting and the normal state. In this experiment, a single crystal lead disk which is a type-I superconductor will be examined.

In addition to the Meissner and the normal state there occurs a combination of the two states, the so called intermediate state, in superconductors with nonzero demagnetization factor N . In this case the intermediate state splits in a superconducting and a nonsuperconducting domain in a range between

$$(1 - N)B_c < B_{ap} < B_c \quad (1)$$

The critical magnetic field B_c is temperature dependent and is given by the following relation [4, 5]

$$B_c(T) = B_c(0) \left(1 - \left(\frac{T}{T_c} \right)^2 \right) \quad (2)$$

With the literature values for the demagnetization factor $N = 0.7595$ [6, 7] and the critical temperature $T_c = 7.19$ K [8] and the relations from equation 1 and equation

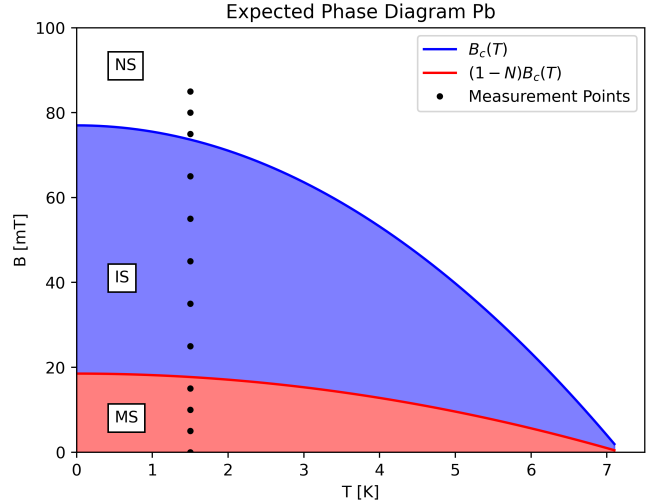


Figure 1: The figure shows the expected phase diagram for Pb with demagnetization factor of $N = 0.7595$ and critical temperature $T_c = 7.19$ K. The red area denotes the Meissner state (MS), the blue area the intermediate state (IS) and the white are the normal state (NS). The black dots are the measured points.

2 one can get the expected phase diagram for the type-I superconductor Pb which is plotted in figure 1. The internal magnetic field inside the sample B_N is expected to be $B_N = B_c$ for $B_{ap} < B_c$ where B_{ap} is the applied external magnetic field and $B_N = B_{ap}$ for $B_{ap} > B_c$ [9]. In this experiment the intermediate state of single crystal lead will be examined and the critical magnetic field will be determined. Furthermore, the magnetic history effects will be examined by measuring once with increasing and once with decreasing magnetic fields.

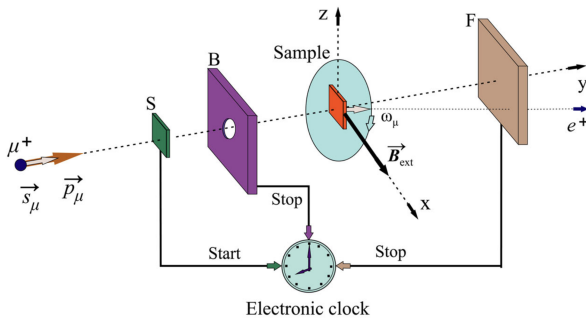


Figure 2: Overview of the experimental setup of a typical μ SR experiment. Spin-polarized muons with spins $\vec{s}_\mu$ parallel to their momenta $\vec{p}_\mu$ are implanted into a sample. This sample is placed between two positron detectors, one facing forward (F) and one facing backward (B). When a muon enters the experiment, a clock is started, and when a decay positron is detected in one of the detectors, the clock is stopped [10]. The actual experimental setup differs from this figure. The detectors are placed above and under the sample, whereas in this figure, the detectors are placed facing forward and backward.

II. THE μ SR TECHNIQUE

In this section we give a brief introduction to the μ SR technique. A muon is a subatomic particle that is similar to an electron, but is about 200 times heavier. It has a spin of $1/2$ and a mean lifetime of about $\tau_\mu \simeq 2.2 \mu\text{s}$. Muons can be positively or negatively charged. The muon spin rotation/relaxation (μ SR) technique primarily uses positive muons, which can be produced with a high degree of polarization. This allows the muons to act as a local magnetic probe when implanted in a solid sample, allowing us to study the magnetic properties of the material. Muons are produced when positive pions decay: $\pi^+ \rightarrow \mu^+ + \nu_\mu$. At PSI, pions are generated by directing a beam of protons with a kinetic energy of 590 MeV at two graphite targets.

The μ SR technique is based on the observation of the time evolution of the muon spin polarization $P(t)$ of an ensemble of muons after their implantation into a sample ($t = 0$). When positive muons decay, they emit a positron, a muon antineutrino, and an electron neutrino $\mu^+ \rightarrow e^+ + \nu_e + \bar{\nu}_\mu$. The positron is preferentially emitted along the direction of the muon spin at the time of the decay (a phenomenon known as decay asymmetry). By measuring the direction and timing of a large number of decay positrons (typically several million per detector), one can directly track the time evolution of the polarization of a group of muons. The experimental setup of a typical μ SR experiment is illustrated in Figure 2.

When a muon is implanted into the sample, a clock is started as the muon passes through the start detector. After the muon quickly thermalizes in the sample, its spin begins to precess at the Larmor frequency $\omega_\mu = \gamma_\mu \cdot B_{\text{loc}}$

($\gamma_\mu = 2\pi \cdot 135.5 \text{ MHz T}^{-1}$ is the gyromagnetic ratio of the muon) in the local magnetic field B_{loc} until it decays and emits a positron. The decay positron emitted by the muon is detected by one of the positron detectors, which stops the clock. This process is repeated for a large number of muons ($\sim 10^6$), and histograms (μ SR time spectra) are generated from the resulting data. The histogram as a function of time for the upwards ($N_U(t)$; U denotes upwards with respect to the initial spin) and the downwards ($N_D(t)$) detectors can be described with the following formula:

$$N_{U(D)}(t) = N_0 \exp(-t/\tau_\mu) (1 \pm A_0 P(t)) \quad (3)$$

A_0 is here the maximum precession amplitude at $t = 0$. This value depends on several factors (such as the detector solid angle and efficiency) and the experimental configuration and it typically is slightly lower than the intrinsic asymmetry of the muon decay, which is $\frac{1}{3}$. The time evolution of the muon's spin polarization can be directly extracted by looking at the so-called asymmetry function, $A(t)$, which is determined as the difference of the signals observed by the two positron detectors, normalized by their sum:

$$A(t) = A_0 P(t) = \frac{N_U(t) - \alpha N_D(t)}{N_U(t) + \alpha N_D(t)} \quad (4)$$

The parameter α takes into account the different solid angles and efficiencies of the positron detectors and is determined through a calibration experiment. [8–10]

III. EXPERIMENTAL PART

A. Experimental Setup

For our experiment a single-crystal lead disk was used. The sample had a purity of 99.999%. The disk was 1.6 mm thick and had a diameter of 10 mm. The measurements were all performed at the General Purpose Surface-Muon Instrument (GPS) at the Paul Scherrer Institute in Villigen, Switzerland at the beam line $\pi\text{M3.2}$. For the cooling of the sample, a ^4He cryostat was used. The cooling process took approximately one hour. The muons had a momentum of $\approx 4 \text{ MeV}/c$. For each data point $\approx 8 \times 10^5$ positron events were taken.

B. Procedure

At first a calibration measurement was done at $T = 8.5 \text{ K}$ where the lead sample is in the normal state. A first measurement was done at zero magnetic field and then the magnetic field was increased from 10 mT to 90 mT in steps of 20 mT. Afterwards, two B-scans were measured, so the temperature was held constant at $T = 1.5 \text{ K}$ and varied the magnetic field. In the first B-scan the magnetic field has been increased (up-scan)

while in the second, B has been decreased (down-scan). From theory, the transition to the intermediate state is expected to happen at a magnetic field of $\approx 18\text{mT}$. The applied fields in the first B-scan were: 0mT, 5mT, 10mT, 15mT, 25mT, 35mT, 45mT, 55mT, 65mT, 75mT, 80mT and 85mT. The analysis of the data showed, that the intermediate state occurs at a higher magnetic field than expected. For the down-scan we measured at the same applied fields in reversed order. The temperature was again kept constant at $T = 1.5\text{K}$ during the second B-scan.

C. Data Analysis

For the evaluation of the data, the MUSRFIT package [11] was used. The obtained μSR -signal was splitted into two different components: the contribution from the sample (s) and the contribution from the background (bg). The background comes from the muons that stopped outside the sample. Therefore, the asymmetry can be written as

$$A(t) = A_s P_s(t) + A_{bg} P_{bg}(t) \quad (5)$$

with the asymmetries A_s, A_{bg} and the muon spin polarisations P_s, P_{bg} that change with time. The background polarisation can be described as

$$P_{bg}(t) = e^{-\frac{1}{2}\sigma_{bg}^2 t^2} \cos(\gamma_\mu B_{ap} t + \phi) \quad (6)$$

with the background depolarisation rate σ_{bg} , the muon gyromagnetic ratio $\gamma_\mu \approx 2\pi \cdot 135.5342\text{MHz T}^{-1}$ [9], the applied magnetic field B_{ap} and the initial phase of the muon spin ensemble ϕ . The polarisation of the sample is then assumed to be separated into a contribution of the normal state and a contribution of the superconducting state which can be described by

$$P_s(t) = f_N e^{-\frac{1}{2}\sigma_n^2 t^2} \cos(\gamma_\mu B_n t + \phi) + (1 - f_N) \left[\frac{1}{3} + \frac{2}{3}(1 - \sigma_{GKT}^2 t^2) e^{-\frac{1}{2}\sigma_{GKT}^2 t^2} \right] \quad (7)$$

with the volume fraction of the normal state f_N , the depolarisation rate σ_n and the internal field B_N . For the superconducting part of the equation, we used the Gaussian Kubo-Toyabe function with the relaxation rate σ_{GKT} (in square brackets) which is normally used to describe the contribution of nuclear magnetic moments in zero-field experiments [12].

Equation 5 was fitted globally over one cycle. The parameters A_{bg}, A_s, σ_{bg} and σ_n were considered as global parameters, whereas the parameters B_{ap}, B_N, f_N and σ_{GKT} were fitted individually for every measurement point. Figures 3 (a)-(c) show the μSR time spectra at $T = 1.5\text{K}$ and $B_{ap} = 80/35/5\text{mT}$. The corresponding Fourier transformations are plotted in figures 3 (d)-(f).

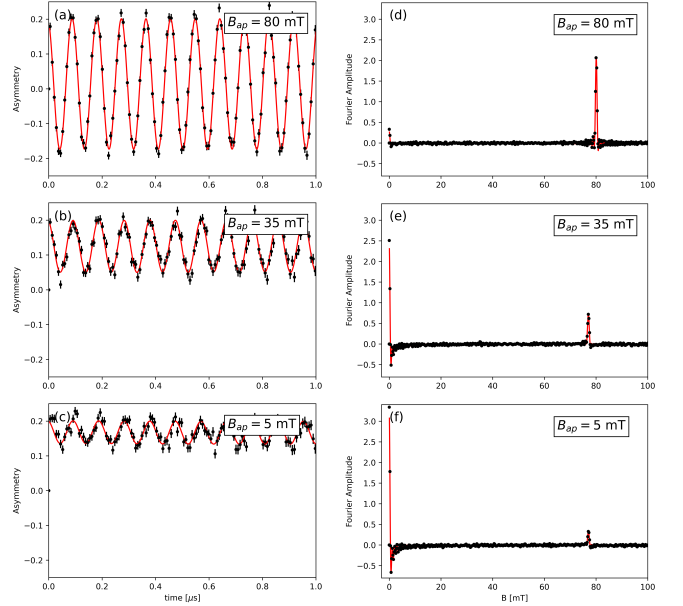


Figure 3: (a)-(c): μSR time spectra with fits at $T = 1.5\text{K}$ and $B = 80\text{mT}$ (a), $B = 35\text{mT}$ (b) and $B = 5\text{mT}$ (c). (d)-(f): Fourier transformation of the time spectra gives us the distribution of the magnetic fields inside the sample.

IV. RESULTS

A. Determination of critical field

The following section presents the experimental data that was collected using measurement procedures previously described. The magnetic fields, at which the measurements were done (while temperature $T = 1.5\text{K}$ was kept fixed) are shown in the phase diagram in Figure 1 (black dots). The time spectra at $B = 5$, $B = 35$ and $B = 80\text{ mT}$ with their Fourier transformation are shown in figure 3. The solid red lines are fits of equation 5 with the background and sample contributions described in equation 6 respectively 7. Two sharp peaks can be seen in the Fourier transformations at $B = 0\text{ mT}$ and at $B = B_c$. At higher applied magnetic fields, the peak at $B = B_c$ is dominating in the Fourier transformation, whereas at lower magnetic fields, the peak at $B = 0\text{ mT}$ is clearly the higher of the two. In Figure 4 the entire magnetic field scan (up and down) is shown. In the intermediate state, both peaks can be seen at $B = 0\text{ mT}$ and at $B = B_c$. From $B_{ap} = 0\text{ mT}$ to $B_{ap} = 85\text{ mT}$ there is a clear upwards trend at $B = B_c$ and a downwards trend at $B = 0\text{ mT}$.

In figure 5 the volume fraction f_N of the normal state is shown as a function of the applied field B_{ap} . f_N starts at zero and remains at that level until a magnetic field of $B_{ap} > (1 - N)B_c$ is applied. As the magnetic field increases from $B_{ap} = (1 - N)B_c$ to B_c , f_N also increases. During this range, the sample is in an intermediate state.

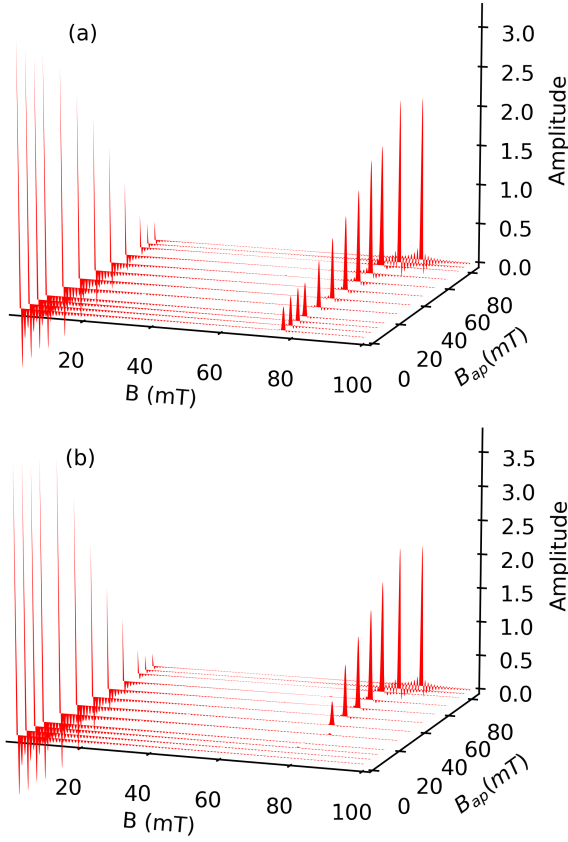


Figure 4: The magnetic field distribution at various applied magnetic fields are shown. The temperature $T = 1.5K$ is held constant (B-scan). In a) the field scan down is shown, whereas in b) the field scan up is shown. For low applied fields, only a peak at $B = 0$ can be seen, but at high applied field, a peak at $B = B_c$ dominates. In the intermediate state, both peaks are clearly visible.

Once the magnetic field reaches B_c , f_N stays constant at a level close to $f_N = 1$, indicating that the sample has entered the normal state. In the down-scan, f_N behaves differently and decreases after reaching B_c , but never drops back to zero. This is due to so called geometrical barrier effects [13]. In various papers [13–16] the main conclusions refer to the existence of topological irreversibility associated to the formation of tubular structures, when the flux enters the material, and laminar patterns in the case of flux exit. In the further analysis, only the upwards field-scan is considered.

In figure 6 the correlation between B_N and the applied field B_{ap} is shown. The function remains constant until $B_{ap} = B_c$ is reached. From there on, B_N has approximately the same value as B_{ap} . From the constant function ($B_{ap} < B_c$) one can determine B_c by fitting a constant function on this section:

$$B_c = (77.16 \pm 0.06) \text{ mT} \quad (8)$$

A similar value can be obtained by calculating the inter-

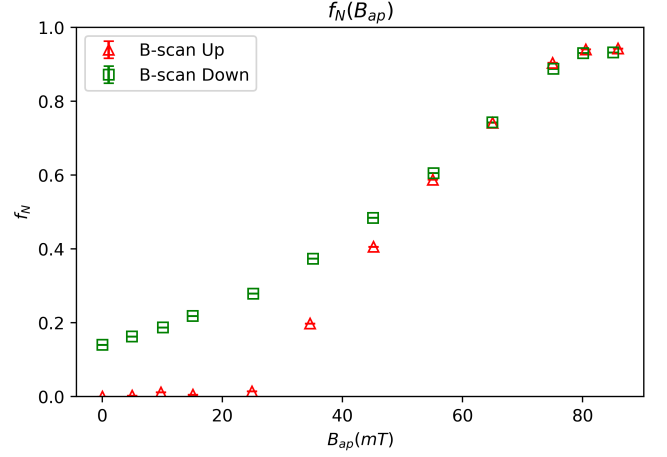


Figure 5: The volume fraction f_N of the normal state as a function of the applied field B_{ap} is shown. f_N remains at zero until a magnetic field of $B_{ap} > (1 - N)B_c$ is applied. From $(1 - N)B_c$ to B_c , f_N increases. In this interval, the sample appears to be in the intermediate state. After $B_{ap} = B_c$ is reached, f_N remains constant close to $f_N = 1$. The sample has therefore reached the normal state. The down-scan shows a different behavior of f_N . f_N decreases after $B_{ap} = B_c$, but never reaches $f_N = 0$.

section between the constant function ($B_{ap} < B_c$) and the linear function ($B_{ap} > B_c$). The value obtained from this calculations is then given by

$$B_c = (77.3 \pm 0.1) \text{ mT} \quad (9)$$

This is in good agreement with the literature value [17]

$$B_{c,\text{lit.}} = (77.17 \pm 0.02) \text{ mT}. \quad (10)$$

B. Determination of the demagnetization factor

The theoretical demagnetization factor N_{th} of a disk with height $2h$ and radius r in an axial magnetic field is given by

$$N_{th}^{-1} = 1 + \frac{1}{q} \frac{h}{r} \quad (11)$$

where q can be written as [6, 7]

$$q = \frac{4}{3\pi} + \frac{2}{3\pi} \tanh \left[1.27 \frac{h}{r} \ln \left(1 + \frac{r}{h} \right) \right] \quad (12)$$

For our sample, this results in a demagnetization factor of

$$N_{th} = 0.7595 \quad (13)$$

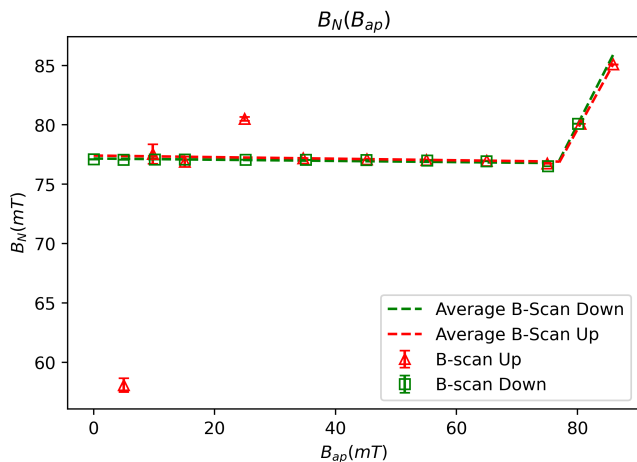


Figure 6: The magnetic field B_N as a function of the applied field B_{ap} is shown. The function remains constant until the critical field B_c is reached. From $B_{ap} = B_c$ the function is linearly rising at a slope of $m = (0.96 \pm 0.05)$. It can be approximately said, that B_N and B_{ap} take the same values after the critical field is reached.

From figure 5 the demagnetization factor N_{exp} can be determined by fitting a linear function to the theoretical intermediate state, one obtains the value

$$(1 - N)B_c = (22.3 \pm 1.8) \text{ mT} \quad (14)$$

Equation 14 then allows the extraction of the demagnetization factor:

$$N_{exp} = (0.71 \pm 0.02) \quad (15)$$

The theoretical value does almost coincide with the experimental one. A possible explanation for the small discrepancy could be small impurities or asymmetries in the sample. It would require further investigation to determine the exact reason for the difference to the literature value.

V. CONCLUSION

In this experiment, a single crystal Pb-disk was used to study the type-I superconductivity with the muon spin rotation/relaxation technique. Type-I superconductors will transition abruptly in a nonsuperconducting state when an external magnetic field is increased to a certain critical field B_c . Since the sample had a nonzero demagnetization factor N , there is an intermediate state between the superconducting Meissner state and the normal state. In this intermediate state the sample is separated in superconducting and nonsuperconducting domains which allows the measurement of the critical field B_c .

The two obtained values for the critical field $B_c = (77.16 \pm 0.06) \text{ mT}$ and $B_c = (77.3 \pm 0.1) \text{ mT}$ are in good agreement with the literature. Furthermore, the demagnetization factor was determined from the magnetic field at the transition to the intermediate state and the critical field. The resulting value of $N_{exp} = (0.71 \pm 0.02)$ is slightly lower than the demagnetization factor calculated from theory $N_{th} = 0.7595$.

Additionally, magnetic history effects were observed when the volume fraction of the normal state f_N was determined for the up-scan and the down-scan. For a decreasing applied field, the volume fraction f_N did not decrease to zero when the sample was in the Meissner state. This can be explained by geometrical barrier effects.

-
- [1] H. K. Onnes, Further experiments with liquid helium. c. on the change of electric resistance of pure metals at very low temperatures etc. v. the disappearance of the resistance of mercury., Comm. Phys. Lab. Univ. Leiden, 122b (1911).
 - [2] W. Meissner and R. Ochsenfeld, Ein neuer effekt bei eintritt der supraleitfähigkeit, *Naturwissenschaften* **21**, 787 (1933).
 - [3] H. K. Onnes, Further experiments with liquid helium. g. on the change of electric resistance of pure metals, etc. vi. on the sudden change in the rate at which the resistance of mercury disappears., Comm. Phys. Lab. Univ. Leiden, 124c (1911).
 - [4] M. Tinkham, *Introduction to Superconductivity*, 2nd ed. (Dover Publications, 2004).
 - [5] C. Poole, H. Farach, R. Creswick, and R. Prozorov, *Superconductivity*, 3rd ed. (Elsevier, Amsterdam, 2014).
 - [6] E. H. Brandt, Geometric edge barrier in the shubnikov phase of type-II superconductors, *Low Temperature Physics* **27**, 723 (2001), <https://doi.org/10.1063/1.1401181>.
 - [7] R. Prozorov and V. G. Kogan, Effective demagnetizing factors of diamagnetic samples of various shapes, *Phys. Rev. Applied* **10**, 014030 (2018).
 - [8] A. Amato, Lecture notes "physics with muons: from atomic physics to solid state physics" (2018).
 - [9] R. Karl, F. Burri, A. Amato, M. Donegà, S. Gvasaliya, H. Luetkens, E. Morenzoni, and R. Khasanov, Muon spin rotation study of type-I superconductivity: Elemental β -sn, *Phys. Rev. B* **99**, 184515 (2019).
 - [10] R. Khasanov, Z. Guguchia, A. Maisuradze, D. Andreica, M. Elender, A. Raselli, Z. Shermadini, T. Goko, F. Knecht, E. Morenzoni, and A. Amato, High pressure research using muons at the paul scherrer institute, *High Pressure Research* **36**, 140 (2016).
 - [11] A. Suter and B. Wojek, Musrfit: A free platform-independent framework for μ sr data analysis, *Physics Procedia* **30**, 69 (2012).

- [12] A. Yaouanc and P. D. de Réotier, *Muon Spin Rotation, Relaxation and Resonance* (Oxford University Press, New York, 2011).
- [13] S. Vélez, C. Panadès-Guinart, G. Abril, A. García-Santiago, J. M. Hernandez, and J. Tejada, Topological magnetic irreversibility in superconducting pb samples of various shapes, *Phys. Rev. B* **78**, 134501 (2008).
- [14] M. Menghini and R. J. Wijngaarden, Patterns close to the critical field in type-i superconductors, *Phys. Rev. B* **75**, 014529 (2007).
- [15] R. Prozorov, Equilibrium topology of the intermediate state in type-i superconductors of different shapes, *Phys. Rev. Lett.* **98**, 257001 (2007).
- [16] R. Prozorov, A. F. Fidler, J. R. Hoberg, and P. C. Canfield, Suprafroth in type-i superconductors, *Nature Physics* **4**, 327 (2008).
- [17] D. L. Decker, D. E. Mapother, and R. W. Shaw, Critical field measurements on superconducting lead isotopes, *Phys. Rev.* **112**, 1888 (1958).